code will have ... subject = "computer". When naming variables we can use underscores but there should be no spaces. Variables are also case sensitive. We cannot use names related to

Python as well like print. Every value has a type each text is put in quotation marks and the type is string. Whole numbers are int and decimals are float. value = "Python" → str ; number = 100 → int ; order = 10.1 → float We can convert between types because when a user inputs Python always enters text/string so we can convert to float or int. number = float(input("Please enter number")). We can also use + to concatenate string in Python. print("20" + "20" + "20") → 202020 we can use f-strings to format our print. print(f"The number is: {number:>10.2f}") → The number is: +20.00. This means the number 20 will be right aligned in 10 spaces, have a sign and 2 decimal places.

NK1

Defining creates a usable block of code. Calling → writing the function's name with () runs it. return sends a value back so it can be stored or used later. Print() displays a value on the screen. Positional values are matched to parameters by their order. Keyword values are matched using parameter names. Default a parameter has a value used when no argument is provided. *args collects any number of positional arguments into a tuple. **kwargs collects any number of keyword arguments into a dictionary. local variable are created inside a function and normally can only be accessed there. Use return to send a result outside a function. A docstring describes what a function does. Write inside triple quotes immediately after the function definition. A module is a python file or library containing reusable code. import lets you use code from another module. if __name__ == "__main__" the block runs when a python file is executed directly, but not when imported as a module.

```
def greet():
    print("Welcome")
greet()
def multiply(a,b):
    return a*b
answer = multiply(3,4)
print(answer)
```

```
def describe(name,age=18):
    print(name,age)
describe("Lina",20) → Positional
describe(age=19,name="Anna") Keyword
describe("Mia") → Default
```

```
def show(*args,**kwargs):
    print(args)
    print(kwargs)
show(4,7,colour="blue",size="large")
```

NK2

A list stores multiple values and can be changed. List comprehension is a quick way to make a new list. Tuples are like a list but are immutable. Dictionaries store information as key:value. Sets store unique values only. Files are used to read and write files. "w" means writes. "r" means read. .strip() removes extra whitespaces and .split() separates text. We use the try/except when handling a code that might ~~case an~~ cause an error.

```
fruits = ["apple","banana","orange"]
fruits.append("mango") # adds mango
fruits.remove("banana") # removes banana
numbers = [1,2,3,4]
doubles = [n*2 for n in numbers]
point = (3,5)
print(point[0]) # 3
students = [
    {"name":"Ali","mark":90},
]
```

```
student = {
    "name": "Ali",
    "age": 20
}
numbers = {1,2,2,3}
print(numbers) # {1,2,3}
with open("data.txt","w") as f:
    f.write("Hello")
```

```
try:
    number = int("hello")
except ValueError:
    print("Not a Number")
```

NK3

beginning of a line of code. While loops false it stops repeating round and go to the item in a sequence can go through. Enumerate both index and item. looping. While loop used

```
value = 40 limit = 50
print(value != limit) = True
print(value >= limit) = false
if age >= 18:
    print("You are an adult")
else:
    print("You are not and
for i in range(3):
    print(i)
```

range(5) → 0,1,2,3,4

range(2,10,2) → 2,

start stop step